Syllabus for CSCI 6050 – Digital Image Analysis and Understanding

Course Name:

Digital Image Analysis and Understanding

Course Prefix and Number:

CSCI 6050

Course Section Number:

001

Course Description (from catalog):

Examines the mathematical foundations of image processing. Includes object detection, recognition, and classification techniques. Coursework introduces students to convolutional neural networks and explores state-of-the-art computer vision approaches.

Required Course Materials for Purchase:

There are no required materials for this course.

Student Learning Outcomes or Objectives:

- Apply image processing techniques to edit photos in useful ways.
- Use matrices to perform rotation, scaling, and perspective transformations on images.
- Define image features that can be used for tasks such as object detection and correspondence.
- Understand classical classification, detection, and segmentation techniques.
- Investigate modern artificial-intelligence algorithms for computer vision.
- Understand the design of Convolutional Neural Networks

Assessment of Student Performance – Grading Scale:

Letter Grade: Percentage Range

A : 90-100
B : 80-89
C : 70-79
F : 0-69

Assessment of Student Performance – Major Assignments and percentage of final grade:

Category: Percentage of Grade

Programming Assignments: 65%
Exams: 25%
Practice Homeworks and Quizzes: 10%

Assessment of Student Performance – Attendance/participation if portion of final grade:

Attendance is not a assigned portion of the final grade.

Statement of Engagement with Diverse Scholarly Perspectives

This course aligns with the intent of the university to engage diverse scholarly perspectives to develop critical thinking, analysis, and debate. The inclusion of a course reading does not imply endorsement.